Sector decomposition for an anisotropic oscillatory integral

1 Setting and overview

1.1 The integral

Let $\ell_1, \dots, \ell_d \geq 0$ and, for $s > 0$, let $A_s = \operatorname{diag}(s^{\ell_1}, \dots, s^{\ell_d})$. For $t \geq 1$ and $\xi \in \mathbb{R}^d$ we define the phase

$$\psi(t, \xi) = \int_1^t |A_s \xi| \, ds.$$

We study the oscillatory integral

$$I(t, x) = \int_{\mathbb{R}^d} e^{i(x \cdot \xi - \psi(t,\xi))} \, |A_t \xi|^{-1} \, \chi(\xi) \, d\xi,$$

where χ is a smooth cutoff supported in the annulus $\{\frac{1}{2} \leq |\xi| \leq 2\}$. We want to bound $I(t, x)$ for large t, uniformly in x.

1.2 The difficulty and the plan

The matrix A_t stretches each frequency coordinate by a different power of t. So there is no single large parameter in front of the phase. We handle this in two moves.

First we split frequency space into d sectors, according to which weighted coordinate $t^{\ell_j}|\xi_j|$ is the largest. On sector k we use the dominant coordinate ξ_k and the $d-1$ ratios $\eta_j = t^{\ell_j} \xi_j / (t^{\ell_k} \xi_k)$ as new variables. In these variables the sector integral becomes an outer integral in ξ_k of an inner oscillatory integral in η with a single large parameter $\lambda = t^{\ell_k + 1} |\xi_k|$ (Section 2).

Second we bound the inner integral by stationary phase. The phase is strictly concave in η (Section 3). Its higher derivatives are not bounded uniformly in t, but they are once we weight them correctly (Section 4). A direct application of the stationary phase theorem gives decay in λ but loses a power of t through the amplitude (Section 5). The loss disappears when the estimate is carried out at the natural scale $\lambda^{-1/2}$. This gives the main result, Theorem 3 (Section 6). The outer integral in ξ_k is not treated here.

1.3 Conventions

In all estimates $t \geq 2$. The index k is fixed, and i, j, m run over $\{1, \dots, d\} \setminus \{k\}$. Multi-indices $\alpha = (\alpha_j)_{j \neq k}$ have $d-1$ entries, and δ_i is the multi-index with a 1 in position i and 0 elsewhere. We write C, c for positive constants that depend only on d and $\ell_1, \dots, \ell_d$ and may change from line to line. C_α may also depend on α.

2 Reduction to one sector

2.1 Sectors

Sector k is the set of ξ for which $t^{\ell_k}|\xi_k|$ is the largest of the weights $t^{\ell_1}|\xi_1|, \dots, t^{\ell_d}|\xi_d|$. Since $|A_t\xi|^2$ is the sum of the d squared weights, on sector k

$$t^{\ell_k}|\xi_k| \;\geq\; \frac{|A_t\xi|}{\sqrt{d}}.$$

Every ξ lies in some sector, so I is a sum of d sector integrals. We fix k.

2.2 New variables

For $j \neq k$ we introduce the ratios

$$\eta_j = \frac{t^{\ell_j}\xi_j}{t^{\ell_k}\xi_k},$$

which compare the weight of coordinate j with the dominant weight. On sector k they satisfy $|\eta_j| \leq 1$. We write $\eta = (\eta_j)_{j\neq k} \in \mathbb{R}^{d-1}$ and use (ξ_k, η) as coordinates on the sector.

2.3 The integrand in the new variables

We rewrite each factor of the integrand in terms of ξ_k and η.

The amplitude. Since

$$|A_t\xi|^2 = t^{2\ell_k}\xi_k^2 + \sum_{j\neq k} t^{2\ell_j}\xi_j^2 = t^{2\ell_k}\xi_k^2\Big(1 + \sum_{j\neq k}\eta_j^2\Big),$$

we have

$$|A_t\xi| = t^{\ell_k}|\xi_k|\,(1+|\eta|^2)^{1/2}, \qquad |A_t\xi|^{-1} = t^{-\ell_k}|\xi_k|^{-1}(1+|\eta|^2)^{-1/2}.$$

The phase ψ. In the same way, for $0 < s \leq t$,

$$|A_s\xi| = s^{\ell_k}|\xi_k|\Big(1 + \sum_{j\neq k}\frac{s^{2\ell_j}\xi_j^2}{s^{2\ell_k}\xi_k^2}\Big)^{1/2}.$$

We measure time in units of t: substitute $s = \sigma t$, so $ds = t\,d\sigma$ and $s^{2(\ell_j-\ell_k)}\xi_j^2/\xi_k^2 = \sigma^{2(\ell_j-\ell_k)}\eta_j^2$. This gives

$$\psi(t,\xi) = t^{\ell_k+1}|\xi_k|\,G(t,\eta), \qquad G(t,\eta) = \int_{1/t}^{1}\sigma^{\ell_k}\Big(1 + \sum_{j\neq k}\sigma^{2(\ell_j-\ell_k)}\eta_j^2\Big)^{1/2}d\sigma.$$

So the phase is a large factor $t^{\ell_k+1}|\xi_k|$ times a function G of η, whose only dependence on t is through the lower limit $1/t$.

The linear term. We aim at a bound uniform in x, so we may replace x by $A_t x$ without changing $\sup_x |I(t,x)|$. The linear part of the phase becomes

$$A_t x\cdot\xi = x_k t^{\ell_k}\xi_k + \sum_{j\neq k} t^{\ell_j}x_j\xi_j = x_k t^{\ell_k}\xi_k + t^{\ell_k}\xi_k\sum_{j\neq k}x_j\eta_j.$$

The measure. For fixed ξ_k, the change of variables $\xi_j = t^{\ell_k-\ell_j}\xi_k\eta_j$, $j \neq k$, gives

$$d\xi = |\xi_k|^{d-1}\prod_{j\neq k} t^{\ell_k-\ell_j}\, d\xi_k\, d\eta = t^{\,d\ell_k-\sum_{j=1}^d \ell_j}\,|\xi_k|^{d-1}\, d\xi_k\, d\eta.$$

2.4 The sector integral

Denote by I_k the part of I coming from sector k. Collecting the formulas above,

$$I_k = t^{\,d\ell_k-\sum_j \ell_j-\ell_k}\int_{\mathbb{R}} e^{ix_k t^{\ell_k}\xi_k}\,|\xi_k|^{d-2}\,J(\xi_k)\,d\xi_k, \qquad J(\xi_k) = \int_{[-1,1]^{d-1}} e^{i\lambda\Phi(\eta)}\,b(\xi_k,\eta)\,d\eta,$$

where the large parameter, the phase and the amplitude are

$$\lambda = t^{\ell_k+1}|\xi_k|, \qquad \Phi(\eta) = \frac{\xi_k}{|\xi_k|}\,\frac{1}{t}\sum_{j\neq k} x_j\eta_j - G(t,\eta),$$

$$b(\xi_k,\eta) = (1+|\eta|^2)^{-1/2}\,\chi\Big(\xi_k e_k + \sum_{j\neq k} t^{\ell_k-\ell_j}\xi_k\eta_j\, e_j\Big).$$

The rest of these notes bounds the inner integral J. Two remarks on its ingredients.

The phase. Φ depends on x and on the sign of ξ_k only through its linear part. All derivatives of order two and higher are those of $-G$ and do not see x at all.

The amplitude and the size of λ. The second factor of b is the cutoff χ in the new variables. Since $|\xi| = |\xi_k|\big(1+\sum_{j\neq k} t^{2(\ell_k-\ell_j)}\eta_j^2\big)^{1/2}$, the amplitude b is supported where

$$\frac{1}{2} \le |\xi_k|\Big(1+\sum_{j\neq k} t^{2(\ell_k-\ell_j)}\eta_j^2\Big)^{1/2} \le 2.$$

With $\ell_{\min} = \min_j \ell_j$ and $|\eta_j| \le 1$, this gives

$$\frac{t^{\ell_{\min}-\ell_k}}{2\sqrt{d}} \;\le\; |\xi_k| \;\le\; 2 \qquad \text{on the support of } b,$$

and therefore $\lambda \ge t^{1+\ell_{\min}}/(2\sqrt{d})$. So λ is indeed large. Note, however, that a derivative ∂_{η_j} falling on the cutoff produces a factor $t^{\ell_k-\ell_j}\xi_k$, which is large when $\ell_j < \ell_k$. This will be the main obstacle in Section 5.

3 The Hessian of the phase

Set

$$H(\sigma,\eta) = \Big(1+\sum_{j\neq k}\sigma^{2(\ell_j-\ell_k)}\eta_j^2\Big)^{1/2}, \qquad \text{so that} \qquad G(t,\eta) = \int_{1/t}^1 \sigma^{\ell_k}\,H(\sigma,\eta)\,d\sigma.$$

The goal of this section is to show that Φ is strictly concave in η, with a Hessian bounded below on compact sets uniformly in t.

First derivatives. Since $\partial_{\eta_j} H = \sigma^{2(\ell_j-\ell_k)}\eta_j\, H^{-1}$,

$$\partial_{\eta_j}\Phi(\eta) = \frac{\xi_k}{|\xi_k|}\,\frac{x_j}{t} - \eta_j \int_{1/t}^{1} \sigma^{2\ell_j-\ell_k}\, H^{-1}\, d\sigma.$$

Second derivatives. The linear part of Φ has zero Hessian, so $\nabla_\eta^2\Phi = -\nabla_\eta^2 G$. Differentiating once more,

$$\partial_{\eta_i}\partial_{\eta_j} G(t,\eta) = \int_{1/t}^{1} \sigma^{\ell_k}\, H^{-3}\Big(\delta_{ij}\,\sigma^{2(\ell_j-\ell_k)}\, H^2 - \sigma^{2(\ell_i+\ell_j-2\ell_k)}\eta_i\eta_j\Big)\, d\sigma.$$

The quadratic form. Let $D(\sigma) = \operatorname{diag}\big(\sigma^{2(\ell_j-\ell_k)}\big)_{j\neq k}$. In matrix form the second derivatives read

$$-\nabla_\eta^2\Phi = \int_{1/t}^{1} \sigma^{\ell_k} H^{-3}\Big(H^2 D - (D\eta)(D\eta)^T\Big)\, d\sigma.$$

Fix $v \in \mathbb{R}^{d-1}$ and put $z = D^{1/2}\eta$, $u = D^{1/2}v$. Then $\langle Dv, v\rangle = |u|^2$, $\langle D\eta, v\rangle = \langle z, u\rangle$, and $|z|^2 = \langle D\eta, \eta\rangle = H^2 - 1$. Hence

$$-\langle \nabla_\eta^2\Phi\, v, v\rangle = \int_{1/t}^{1} \sigma^{\ell_k} H^{-3}\Big(H^2|u|^2 - \langle z, u\rangle^2\Big)\, d\sigma.$$

By Cauchy–Schwarz, $\langle z, u\rangle^2 \le |z|^2|u|^2 = (H^2-1)|u|^2$, so the bracket lies between $|u|^2$ and $H^2|u|^2$. Writing

$$S(\sigma, v) = |u|^2 = \sum_{j\neq k} \sigma^{2(\ell_j-\ell_k)} v_j^2,$$

we arrive at the two-sided bound

$$\int_{1/t}^{1} \sigma^{\ell_k} H^{-3}\, S\, d\sigma \;\le\; -\langle \nabla_\eta^2\Phi\, v, v\rangle \;\le\; \int_{1/t}^{1} \sigma^{\ell_k} H^{-1}\, S\, d\sigma.$$

Strict concavity. For $v \neq 0$ the left side is positive. So $-\nabla_\eta^2\Phi$ is positive definite and Φ is strictly concave in η, for every t, x and ξ_k.

A lower bound on compact sets. Later we need a lower bound $-\nabla_\eta^2\Phi \ge cI$ on a ball $|\eta| \le R$, with c independent of t. Since the integrand is nonnegative and $t \ge 2$, we may restrict the integral to $\sigma \in [\frac{1}{2}, 1]$. There σ^{ℓ_k} and each $\sigma^{2(\ell_j-\ell_k)}$ lie between two positive constants, so $S \ge c|v|^2$, and $H \le C_R$ when $|\eta| \le R$. Hence

$$-\langle \nabla_\eta^2\Phi(\eta)\, v, v\rangle \;\ge\; \int_{1/2}^{1} \sigma^{\ell_k} H^{-3}\, S\, d\sigma \;\ge\; c_R\, |v|^2 \qquad (|\eta| \le R),$$

with $c_R > 0$ depending only on R, d and the ℓ_j.

4 Higher derivatives of the phase

Let $|\alpha| \ge 2$. Two derivatives kill the linear part of Φ, so $\partial_\eta^\alpha\Phi = -\partial_\eta^\alpha G$, and it suffices to bound $\partial_\eta^\alpha G$. We first give a direct bound. Then we explain why it is not uniform in t, and we prove a weighted bound that is.

4.1 A direct bound

Reduction to one function. For fixed σ put $y_j = \sigma^{\ell_j - \ell_k}\eta_j$, $j \neq k$, and $\langle y\rangle = (1+|y|^2)^{1/2}$. Then $H(\sigma,\eta) = \langle y\rangle$. Since each y_j is a multiple of η_j alone, the chain rule gives

$$\partial_\eta^\alpha H = \prod_{j\neq k} \sigma^{\alpha_j(\ell_j-\ell_k)}\,\big(\partial_y^\alpha \langle y\rangle\big).$$

Derivatives of $\langle y\rangle$. We claim that for $|\alpha|\geq 1$, $\partial_y^\alpha\langle y\rangle$ is a finite linear combination of terms

$$y^\beta\,\langle y\rangle^{1-2m} \qquad \text{with} \qquad m\geq 1,\quad 2m-|\beta| = |\alpha|.$$

For $|\alpha|=1$ this holds, since $\partial_{y_i}\langle y\rangle = y_i\langle y\rangle^{-1}$. Differentiating $y^\beta\langle y\rangle^{1-2m}$ in y_i gives two terms: $\beta_i\, y^{\beta-\delta_i}\langle y\rangle^{1-2m}$, which vanishes if $\beta_i = 0$, and $(1-2m)\,y^{\beta+\delta_i}\langle y\rangle^{1-2(m+1)}$. In both terms $2m-|\beta|$ increases by one, which proves the claim. Since $|y^\beta|\leq \langle y\rangle^{|\beta|}$, each term is bounded by $\langle y\rangle^{|\beta|+1-2m} = \langle y\rangle^{1-|\alpha|}$, so

$$\big|\partial_y^\alpha\langle y\rangle\big| \leq C_\alpha\,\langle y\rangle^{1-|\alpha|}.$$

In words, $\langle y\rangle$ behaves like a function homogeneous of degree one.

The bound. Differentiating under the integral sign and using $\langle y\rangle = H\geq 1$, we obtain for $|\alpha|\geq 1$

$$\big|\partial_\eta^\alpha G(t,\eta)\big| \leq C_\alpha \int_{1/t}^1 \sigma^{\,\ell_k+\sum_{j\neq k}\alpha_j(\ell_j-\ell_k)}\,H(\sigma,\eta)^{1-|\alpha|}\,d\sigma \leq C_\alpha\int_{1/t}^1 \sigma^{\,\ell_k+\sum_{j\neq k}\alpha_j(\ell_j-\ell_k)}\,d\sigma.$$

The last integral stays bounded as $t\to\infty$ only if the exponent of σ is larger than -1. When some $\ell_j<\ell_k$, this fails for large α_j. We look at this more closely.

4.2 Good, bad and oscillatory directions

For $j\neq k$ we call the direction j

- *good* if $\ell_j\geq\ell_k$, and *bad* if $\ell_j<\ell_k$;
- *oscillatory* if $2\ell_j-\ell_k+1>0$.

We write $\mathcal{O}$ for the set of oscillatory directions. Every good direction is oscillatory, since then $2\ell_j-\ell_k+1\geq \ell_k+1>0$. So a non-oscillatory direction is bad. We define the weights

$$v_j(\eta) = 1 \;\;(j \text{ good}), \qquad v_j(\eta) = |\eta_j| \;\;(j \text{ bad}).$$

Why the direct bound fails. High derivatives in a bad direction blow up at $\eta_j=0$. Indeed, set all η_m with $m\neq j$ to zero and expand $H = (1+\sigma^{2(\ell_j-\ell_k)}\eta_j^2)^{1/2}$ in powers of η_j. The coefficient of η_j^{2N} is $\binom{1/2}{N}\sigma^{2N(\ell_j-\ell_k)}$, so

$$\partial_{\eta_j}^{2N} G(t,0) = (2N)!\binom{1/2}{N}\int_{1/t}^1 \sigma^{\,\ell_k+2N(\ell_j-\ell_k)}\,d\sigma.$$

Since $\ell_j<\ell_k$, the exponent is below -1 for N large, and the integral tends to infinity as $t\to\infty$. The reason is a crossover: near $\eta_j=0$ the coordinate j dominates H at early times $\sigma\sim 1/t$, where $\sigma^{\ell_j-\ell_k}$ is huge. Away from $\eta_j=0$ this does not happen, and the weight $v_j=|\eta_j|$ measures how far away we are.

4.3 The weighted bound

Derivatives of H, weighted. Write $\mu_m = \sigma^{\ell_m - \ell_k}$, so that $y_m = \mu_m \eta_m$. The chain rule and the bound for $\langle y \rangle$ give

$$\left|\partial_\eta^\alpha H\right| \le C_\alpha \, \mu^\alpha \, H^{1-|\alpha|} = C_\alpha \, H \prod_{m \neq k} \left(\frac{\mu_m}{H}\right)^{\alpha_m}, \qquad \mu^\alpha = \prod_{m \neq k} \mu_m^{\alpha_m}.$$

Two simple facts control each factor μ_m / H:

- if m is good, then $\mu_m \le 1 \le H$, so $\mu_m / H \le 1$;
- if m is bad, then $H \ge \mu_m |\eta_m|$, so $\mu_m / H \le |\eta_m|^{-1}$.

In both cases $\mu_m / H \le v_m^{-1}$.

Lemma 1. *Let $|\alpha| \ge 2$ with $\alpha_m = 0$ for $m \notin \mathcal{O}$, and choose $i, j \in \mathcal{O}$ (possibly $i = j$) such that $\alpha' = \alpha - \delta_i - \delta_j$ has nonnegative entries. Then for $t \ge 2$,*

$$\left|\partial_\eta^\alpha G(t, \eta)\right| \le C_\alpha \prod_{m \neq k} v_m(\eta)^{-\alpha'_m}.$$

In words: two derivatives in oscillatory directions are free, and each further derivative in a bad direction costs a factor $1/|\eta_m|$. Further derivatives in good directions cost nothing.

Proof. Since $|\alpha| = |\alpha'| + 2$, we can split the bound for $\partial_\eta^\alpha H$ as

$$\mu^\alpha H^{1-|\alpha|} = \frac{\mu_i \mu_j}{H} \prod_{m \neq k} \left(\frac{\mu_m}{H}\right)^{\alpha'_m} \le \frac{\mu_i \mu_j}{H} \prod_{m \neq k} v_m^{-\alpha'_m}.$$

The product on the right does not depend on σ. It remains to integrate the first factor against $\sigma^{\ell_k} d\sigma$. Using $H \ge 1$ and $\mu_i \mu_j \le \frac{1}{2}(\mu_i^2 + \mu_j^2)$,

$$\int_{1/t}^1 \sigma^{\ell_k} \, \frac{\mu_i \mu_j}{H} \, d\sigma \le \frac{1}{2} \int_0^1 \left(\sigma^{2\ell_i - \ell_k} + \sigma^{2\ell_j - \ell_k}\right) d\sigma < \infty,$$

because $i, j \in \mathcal{O}$ means $2\ell_i - \ell_k > -1$ and $2\ell_j - \ell_k > -1$. □

4.4 What the weights suggest

The natural scale of a stationary phase integral with parameter λ is $\lambda^{-1/2}$. Zoom to this scale by setting $\omega = \lambda^{1/2} \eta$. Then

$$\partial_\omega^\alpha (\lambda G) = \lambda^{1-|\alpha|/2} \, \partial_\eta^\alpha G = \lambda^{-|\alpha'|/2} \, \partial_\eta^\alpha G,$$

and Lemma 1 gives

$$\left|\partial_\omega^\alpha (\lambda G)\right| \le C_\alpha \prod_{m \neq k} \left(\lambda^{1/2} v_m\right)^{-\alpha'_m}.$$

For good m we have $\lambda^{1/2} v_m = \lambda^{1/2} \ge 1$, and for bad m we have $\lambda^{1/2} v_m = |\omega_m|$. So all ω-derivatives of λG are bounded as long as $|\omega_m| \ge 1$ in the bad directions, that is, $|\eta_m| \ge \lambda^{-1/2}$. This suggests the following splitting of the η integral, which is carried out in Section 6:

- on the strips $|\eta_m| \le \lambda^{-1/2}$, m bad, we integrate trivially; a strip has width $\lambda^{-1/2}$, which is exactly the gain oscillation would give;
- on the rest, the phase λG is convex with bounded derivatives in ω, so the ω integral is bounded and each oscillatory direction gives the factor $\lambda^{-1/2}$;
- directions outside $\mathcal{O}$ are never differentiated and are integrated trivially.

5 The stationary phase theorem

5.1 Smooth sectors

The stationary phase estimate below needs a smooth amplitude, and the indicator of a sector is not smooth. So we replace the sharp sectors by a smooth partition. Fix smooth $\rho_1, \dots, \rho_d$ on $\mathbb{S}^{d-1}$ with $\sum_k \rho_k = 1$ and $\rho_k(\theta) = 0$ unless $|\theta_k| \geq \frac{1}{2\sqrt{d}}$. Such a partition exists since $\max_k |\theta_k| \geq 1/\sqrt{d}$ on $\mathbb{S}^{d-1}$. We insert $\rho_k(A_t\xi/|A_t\xi|)$ into I_k in place of the indicator of sector k. The η integral then runs over $\mathbb{R}^{d-1}$, with amplitude

$$b(\xi_k, \eta) = (1+|\eta|^2)^{-1/2}\, \tilde{\chi}(\xi_k, \eta)\, \rho_k\Big(\frac{\xi_k}{|\xi_k|}\, \frac{e_k + \sum_{j\neq k} \eta_j e_j}{(1+|\eta|^2)^{1/2}}\Big),$$

where $\tilde{\chi}(\xi_k, \eta) = \chi\big(\xi_k e_k + \sum_{j\neq k} t^{\ell_k - \ell_j} \xi_k \eta_j\, e_j\big)$ is the cutoff in the new variables. The kth component of the argument of ρ_k is $\pm(1+|\eta|^2)^{-1/2}$, so $b(\xi_k, \cdot)$ is smooth and supported in $|\eta| \leq 2\sqrt{d}$. The bounds of Section 2.4 on the support remain true with other constants: $|\xi_k| \leq 2$ and $\lambda \geq c\, t^{1+\ell_{\min}}$.

5.2 The theorem

For $b \in C_0^\infty(\mathbb{R}^n)$ with support K, let K_1 be the set of points at sup distance at most 1 from K, and set

$$M_m = \sum_{2\leq|\alpha|\leq m} \sup_{K_1} |\partial^\alpha \Phi|, \qquad N_m = \sum_{|\alpha|\leq m} \sup_{K} |\partial^\alpha b|, \qquad a_0 = \inf_{K_1} \big|\det \nabla^2 \Phi\big|.$$

Theorem 2 (Alazard, Burq, Zuily, Theorem 3)**.** *Let Φ be real and smooth near K_1, with $a_0 > 0$ and $\nabla\Phi$ injective on K_1. Then for all $\lambda \geq 1$,*

$$\Big| \int_{\mathbb{R}^n} e^{i\lambda\Phi(\eta)}\, b(\eta)\, d\eta \Big| \leq \frac{C_n}{a_0} \big(1 + M_{n+2}\big)^{n/2}\, N_{n+1}\, \lambda^{-n/2},$$

where C_n depends only on n.

Two features of this bound matter for us. It involves no volume of K, so a large support costs nothing. And it sees Φ only through derivatives of order at least two, so the linear part of Φ, which carries x, plays no role.

5.3 A first application, and what it loses

We apply Theorem 2 with $n = d-1$, for fixed t, x, ξ_k. Then $K_1 \subset [-R, R]^{d-1}$ with $R = 2\sqrt{d} + 1$. We check the hypotheses one by one.

- *Injectivity.* $\nabla^2_\eta \Phi$ is negative definite on $\mathbb{R}^{d-1}$, so for $\eta \neq \eta'$,

$$\big\langle \nabla\Phi(\eta) - \nabla\Phi(\eta'), \eta - \eta' \big\rangle = \int_0^1 \big\langle \nabla^2_\eta \Phi\big(\eta' + s(\eta - \eta')\big)(\eta - \eta'), \eta - \eta' \big\rangle\, ds < 0.$$

  Hence $\nabla\Phi$ is injective on all of $\mathbb{R}^{d-1}$, and no localization is needed.

- a_0. By the lower bound of Section 3, all eigenvalues of $-\nabla^2_\eta \Phi$ are at least c_R on K_1, so $a_0 \geq c_R^{d-1}$.

- M_{d+1}. For fixed t the phase is smooth, so M_{d+1} is finite. But Section 4 controls it uniformly in t only in part: the direct bound grows with t in bad directions, and Lemma 1 degenerates near the hyperplanes $\eta_m = 0$, m bad.
- N_d. The factors $(1+|\eta|^2)^{-1/2}$ and $\rho_k(\cdots)$ are smooth in η and independent of t. Each ∂_{η_j} falling on $\tilde{\chi}$ produces a factor $t^{\ell_k-\ell_j}\xi_k$, and $|\xi_k| \le 2$, so
$$N_d \le C\, t^{\,d(\ell_k-\ell_{\min})}.$$
- $\lambda \ge 1$. On the support of b, $\lambda \ge c\, t^{1+\ell_{\min}}$, which is ≥ 1 for t large.

Conclusion of the first application. For t large,
$$\Big|\int_{\mathbb{R}^{d-1}} e^{i\lambda\Phi(\eta)}\, b(\xi_k,\eta)\, d\eta\Big| \le \frac{C}{a_0}\,\big(1+M_{d+1}\big)^{\frac{d-1}{2}}\, t^{\,d(\ell_k-\ell_{\min})}\,\big(t^{\ell_k+1}|\xi_k|\big)^{-\frac{d-1}{2}}.$$
None of a_0, M_{d+1}, N_d depends on x, so the bound is uniform in x. But it is not satisfactory in t: the factor $t^{d(\ell_k-\ell_{\min})}$ may cancel the decay $\lambda^{-(d-1)/2}$, and M_{d+1} is not under control uniformly in t.

Where the loss comes from. Both problems have the same origin. The quantities M and N measure derivatives at unit scale in η, but the stationary phase integral lives at scale $\lambda^{-1/2}$. At that scale the picture is different. A derivative of $\tilde{\chi}$ in direction j costs $t^{\ell_k-\ell_j}|\xi_k|$ per unit of η_j, hence
$$\lambda^{-1/2}\, t^{\ell_k-\ell_j}|\xi_k| = t^{-(2\ell_j-\ell_k+1)/2}\,|\xi_k|^{1/2}$$
per unit of $\omega_j = \lambda^{1/2}\eta_j$, and this is bounded exactly when j is oscillatory. And the strips $|\eta_m| \le \lambda^{-1/2}$, where Lemma 1 degenerates, are so thin that they can be integrated trivially. The next section turns this into a proof.

6 Proof of the main estimate

6.1 Three groups of directions

For $j \ne k$ we distinguish

- $\mathcal{O}$, the oscillatory directions: $2\ell_j - \ell_k + 1 > 0$;
- $\mathcal{T}$, the non-oscillatory directions: $2\ell_j - \ell_k + 1 \le 0$;
- $\mathcal{B} \subset \mathcal{O}$, the bad oscillatory directions: $\ell_j < \ell_k$ and $j \in \mathcal{O}$.

Thus $\mathcal{O} \setminus \mathcal{B}$ consists of the good directions, and every direction in $\mathcal{T}$ is bad. The directions in $\mathcal{O}$ will be treated by stationary phase, those in $\mathcal{T}$ by the size of their support.

6.2 Statement

Recall from Section 5 that $b(\xi_k,\cdot)$ is smooth and supported in $|\eta| \le 2\sqrt{d}$, and that $|\xi_k| \le 2$ and $\lambda \ge c\,t^{1+\ell_{\min}}$ whenever $b(\xi_k,\cdot)$ is not identically zero.

Theorem 3 (Main estimate)**.** *There are t_0 and C, depending only on d and $\ell_1,\dots,\ell_d$, such that for $t \ge t_0$, all $x \in \mathbb{R}^d$ and all $\xi_k \ne 0$,*
$$\Big|\int_{\mathbb{R}^{d-1}} e^{i\lambda\Phi(\eta)}\, b(\xi_k,\eta)\, d\eta\Big| \le C\,\lambda^{-\frac{d-1}{2}} \prod_{j\in\mathcal{T}} t^{\frac{2\ell_j-\ell_k+1}{2}}\,|\xi_k|^{-1/2}, \qquad \lambda = t^{\ell_k+1}|\xi_k|.$$

Each factor in the product over $\mathcal{T}$ is at most $|\xi_k|^{-1/2}$, since $2\ell_j - \ell_k + 1 \le 0$ there. When $\mathcal{T}$ is empty the estimate is the full stationary phase rate $\lambda^{-(d-1)/2}$, with no loss from the amplitude.

Remark. In the $\mathcal{T}$ directions the gain is $\lambda^{-1/2}$ only up to a factor $|\xi_k|^{-1/2}$. This factor has to be handled in the outer ξ_k integral of I_k.

6.3 Strategy

Set $n = d - 1$ and write J for the integral in Theorem 3, with t, x, ξ_k fixed. If $b(\xi_k, \cdot) \equiv 0$ there is nothing to prove, so we assume $|\xi_k| \le 2$ and $\lambda \ge c\, t^{1+\ell_{\min}}$, and we take t_0 so large that $\lambda \ge 1$. The proof has five steps.

1. Zoom to $\omega = \lambda^{1/2}\eta$. This puts the decay $\lambda^{-n/2}$ in front and leaves an integral that only has to be bounded.
2. In each bad oscillatory direction, split the integral into a strip $|\omega_m| \le 3$ and its complement.
3. On each piece, apply Theorem 2 with $\lambda = 1$ in the oscillatory directions that are not in a strip. Lemma 1 gives bounded phase derivatives there, and the oscillatory condition gives bounded amplitude derivatives.
4. Integrate trivially in the remaining directions: strips have bounded width, and non-oscillatory directions have thin support.
5. Add up the pieces.

6.4 Step 1: zoom

Put $\eta = \lambda^{-1/2}\omega$. Then $d\eta = \lambda^{-n/2} d\omega$ and

$$J = \lambda^{-n/2} \int_{\mathbb{R}^n} e^{i\tilde{\Phi}(\omega)}\, \tilde{b}(\omega)\, d\omega, \qquad \tilde{\Phi}(\omega) = \lambda\, \Phi(\lambda^{-1/2}\omega), \quad \tilde{b}(\omega) = b(\xi_k, \lambda^{-1/2}\omega).$$

By the chain rule,

$$\partial_\omega^\alpha \tilde{\Phi}(\omega) = \lambda^{1-|\alpha|/2}\, (\partial_\eta^\alpha \Phi)(\lambda^{-1/2}\omega).$$

For $|\alpha| = 2$ the power of λ is zero, so the Hessian of $\tilde{\Phi}$ in ω equals the Hessian of Φ in η at the corresponding point. All the decay $\lambda^{-n/2}$ now sits in front. It remains to bound the ω integral by a constant, which is Theorem 2 with $\lambda = 1$.

6.5 Step 2: strip cutoffs in the bad directions

Take $\psi \in C_0^\infty(\mathbb{R})$ with $\psi = 1$ on $[-2, 2]$ and $\psi = 0$ outside $[-3, 3]$. For each $m \in \mathcal{B}$ write $1 = \psi(\omega_m) + (1 - \psi(\omega_m))$ and expand the product over $m \in \mathcal{B}$. This splits J into $2^{|\mathcal{B}|}$ pieces J_S, one for each subset $S \subset \mathcal{B}$ (the directions placed in strips):

$$J_S = \lambda^{-n/2} \int_{\mathbb{R}^n} e^{i\tilde{\Phi}}\, \tilde{b} \prod_{m \in S} \psi(\omega_m) \prod_{m \in \mathcal{B}\setminus S} \big(1 - \psi(\omega_m)\big)\, d\omega.$$

In J_S we call $A = \mathcal{O} \setminus S$ the *active* directions. We integrate first in ω_A, with the *frozen* variables $\omega_{S\cup\mathcal{T}}$ held fixed, and then in the frozen variables.

6.6 Step 3: stationary phase in the active variables

Fix S and the frozen variables $\omega_{S\cup\mathcal{T}}$, and consider the inner integral in $\omega_A \in \mathbb{R}^{|A|}$. We apply Theorem 2 in dimension $|A|$, with $\lambda = 1$, with phase $\tilde{\Phi}$ viewed as a function of ω_A, and with amplitude $\tilde{b}$ times the cutoffs. The theorem gives the bound $\frac{C}{a_0}(1+M)^{|A|/2}N$ with no factor of the volume of the support, so the support of size $\lambda^{1/2}$ in ω costs nothing. We check the hypotheses.

(a) The determinant a_0. The support of $\tilde{b}$ lies in $|\eta| \le 2\sqrt{d}$. Its sup-distance-1 neighborhood $\tilde{K}_1$ in ω is a sup-distance-$\lambda^{-1/2}$ neighborhood in η, and $\lambda^{-1/2} \le 1$, so $\tilde{K}_1$ lies in the ball $|\eta| \le 3\sqrt{d}$. There the lower bound of Section 3 gives $-\nabla_\eta^2\Phi \ge cI$, hence $-\nabla_\omega^2\tilde{\Phi} \ge cI$ by Step 1. The Hessian in the active variables is a principal submatrix, and a principal submatrix of a matrix $\ge cI$ is again $\ge cI$. Hence $a_0 \ge c^{|A|}$.

(b) Injectivity. $\tilde{\Phi}$ is strictly concave in ω_A on all of $\mathbb{R}^{|A|}$, so $\nabla_{\omega_A}\tilde{\Phi}$ is injective by the monotonicity argument of Section 5.

(c) The phase derivatives $\tilde{M}$. Let $|\alpha| \ge 2$ with α supported in $A \subset \mathcal{O}$. Then $\partial^\alpha\Phi = -\partial^\alpha G$, and Step 1 together with Lemma 1 gives

$$|\partial_\omega^\alpha\tilde{\Phi}| = \lambda^{1-|\alpha|/2}|\partial_\eta^\alpha G| \le C\prod_m \left(\lambda^{1/2}v_m\right)^{-\alpha'_m}.$$

For good m, $\lambda^{1/2}v_m = \lambda^{1/2} \ge 1$. For bad $m \in A$, that is $m \in \mathcal{B}\setminus S$, we have $\lambda^{1/2}v_m = |\omega_m|$, and $|\omega_m| \ge 1$ on $\tilde{K}_1$: the factor $1-\psi(\omega_m)$ vanishes for $|\omega_m| \le 2$, and $\tilde{K}_1$ extends the support by sup distance 1. So every factor is at most 1, and $\tilde{M}_{|A|+2} \le C$.

(d) The amplitude derivatives $\tilde{N}$. One ω_j derivative is $\lambda^{-1/2}\partial_{\eta_j}$. Falling on $(1+|\eta|^2)^{-1/2}$ or on ρ_k, it costs at most $C\lambda^{-1/2} \le C$. Falling on $\tilde{\chi}$, it produces the factor $t^{\ell_k-\ell_j}\xi_k$, so it costs

$$\lambda^{-1/2}\,t^{\ell_k-\ell_j}|\xi_k| = t^{-(2\ell_j-\ell_k+1)/2}\,|\xi_k|^{1/2} \le \sqrt{2} \qquad (j\in\mathcal{O},\ |\xi_k|\le 2).$$

This is where the oscillatory condition is used: it makes the exponent of t negative. Derivatives of the strip cutoffs are $O(1)$. Hence $\tilde{N}_{|A|+1} \le C$.

Conclusion of Step 3. For every S and every value of the frozen variables $\omega_{S\cup\mathcal{T}}$,

$$\Big|\int_{\mathbb{R}^{|A|}} e^{i\tilde{\Phi}}\,\tilde{b}\prod_{m\in S}\psi(\omega_m)\prod_{m\in\mathcal{B}\setminus S}\big(1-\psi(\omega_m)\big)\,d\omega_A\Big| \le C.$$

6.7 Step 4: trivial integration in the frozen variables

It remains to integrate the constant C over the frozen variables, which are confined to a box.

- $m \in S$. The cutoff $\psi(\omega_m)$ is supported in $|\omega_m| \le 3$, so this costs a factor 6.
- $j \in \mathcal{T}$. On the support of $\tilde{\chi}$ we have $|\xi_j| \le 2$ with $\xi_j = t^{\ell_k-\ell_j}\xi_k\eta_j$, so $|\eta_j| \le 2t^{\ell_j-\ell_k}/|\xi_k|$. In the variable $\omega_j = \lambda^{1/2}\eta_j$ this means $|\omega_j| \le W_j$ with

  $$W_j = \lambda^{1/2}\cdot\frac{2t^{\ell_j-\ell_k}}{|\xi_k|} = 2\,t^{\frac{2\ell_j-\ell_k+1}{2}}\,|\xi_k|^{-1/2} \le 2\,|\xi_k|^{-1/2} \qquad (j\in\mathcal{T}),$$

  so this costs a factor $2W_j$. Here the non-oscillatory condition $2\ell_j - \ell_k + 1 \le 0$ is used: the support in ω_j is no wider than a constant times $|\xi_k|^{-1/2}$.

6.8 Step 5: conclusion

Combining Steps 3 and 4, each piece satisfies

$$|J_S| \le C\,\lambda^{-n/2}\,6^{|S|}\prod_{j\in\mathcal{T}} 2W_j \le C\,\lambda^{-\frac{d-1}{2}}\prod_{j\in\mathcal{T}} t^{\frac{2\ell_j-\ell_k+1}{2}}\,|\xi_k|^{-1/2}.$$

Summing over the $2^{|\mathcal{B}|}$ subsets S proves Theorem 3. The bound is uniform in x because x enters Φ only through its linear part, which none of a_0, $\tilde{M}$, $\tilde{N}$ sees.

Summary. After zooming to $\omega = \lambda^{1/2}\eta$, the oscillatory condition $2\ell_j - \ell_k + 1 > 0$ is exactly what keeps the amplitude smooth at scale 1. Every direction where it fails has support thinner than the stationary phase scale, so it needs no oscillation. The bad oscillatory directions are handled by cutting out a strip of unit width in ω, where the weighted bounds degenerate; integrating over the strip costs the same factor that oscillation would gain.